Cote 140-2

Scans 341–350

[scan 341]

Of course, this character will again be called, "multiplier", and|or "cyclotomic character". [d1] If our fundamental conjecture is true, this character identifies $(\Gamma_\Sigma)_{\mathrm{ab}}$ with $\widehat{\mathbb{Z}}^*$.

On the other hand, if $I = I(\pi)$, the canonical homomorphism

$$\widehat{\widehat{\mathcal{T}}} \longrightarrow \mathfrak{S}_I | \mathcal{S}_I$$

is [d2] not trivial on Σ, but induces a surjective homomorphism

$$\Sigma \longrightarrow \mathfrak{S}_I$$

whence a subgroup $\Sigma^!$ such that

$$\Sigma/\Sigma^! \xrightarrow{\sim} \mathfrak{S}_I.$$

We see at once that $\forall\ \gamma|g \in \widehat{\widehat{\mathcal{T}}}$ which normalizes Σ normalizes $\Sigma^! = \Sigma \cap \widehat{\widehat{\mathcal{T}}}^!$, whence|i.e. $\Sigma^!$ is also invariant in $\mathcal{N}_\Sigma$. [d4] Let $\mathcal{N}^!_\Sigma = \mathcal{N}_\Sigma \cap \widehat{\widehat{\mathcal{T}}}^!$, we have a Cartesian diagram of subgroups of $\widehat{\widehat{\mathcal{T}}}$ morally, $\Sigma^! \simeq \widehat{\mathcal{T}}^{!+}$

$$\begin{array}{ccc} \mathcal{N}^! & \xhookrightarrow{\mathfrak{S}_I} & \mathcal{N} \\ {\scriptstyle \Gamma_\Sigma}\uparrow & & \uparrow{\scriptstyle \Gamma_\Sigma} \\ \Sigma^! & \xhookrightarrow[\mathfrak{S}_I]{} & \Sigma \end{array}$$

giving an injective homomorphism

$$\mathcal{N}/\Sigma^! \longrightarrow \underbrace{\mathcal{N}/\mathcal{N}^!}_{\mathfrak{S}_I} \times \underbrace{\mathcal{N}/\Sigma}_{\Gamma_\Sigma}$$

of which|when we see at once that it is bijective [m2]

Marginal note. [m2] and|for $\Sigma/\Sigma^! \xrightarrow{\sim} \mathcal{N}/\mathcal{N}^!$ $(\simeq \mathfrak{S}_I)$

$$\mathcal{N}/\Sigma^! \xrightarrow{\sim} \mathcal{N}/\mathcal{N}^! \times \mathcal{N}/\Sigma \simeq \mathfrak{S}_I \times \Gamma_\Sigma$$

and consequently we also have

$$\mathcal{N}^!/\Sigma^! \xrightarrow{\sim} \mathcal{N}/\Sigma = \Gamma_\Sigma,$$

i.e. the group Γ_Σ of arithmetically outer automorphisms of (π, Σ) can also be described via the automorphisms [p. 340]

Apparatus for this scan

`140-2-s341-D001` *deletion. Manuscript:* It
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

`140-2-s341-D002` *deletion. Manuscript:* [illeg. 1]
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

`140-2-s341-D003` *deletion. Manuscript:* !

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s341-D004 *deletion. Manuscript:* [illeg. 3]
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

[scan 342]

[p. 173]

that are outer and induce the identity on I.

In the case $(g,\nu)=(0,3)$, we have $\Sigma^!=(1)$, $\Rightarrow$ hence $\Sigma \xrightarrow{\sim} \mathfrak{S}_3$. $\mathcal{N}$ is the d1 normalizer of $\mathfrak{S}_3$ in $\widehat{\widehat{\mathcal{T}}}$, $\mathcal{N}^! \xrightarrow{\sim} \Gamma_\Sigma$ i1 is its centralizer, and $\mathcal{N}$ is identified with the direct product of the two. d2

Now let $I|I' \subset I$, $\nu' = \operatorname{card} I'$, i2 and let us consider the outer group π' obtained from π by "plugging the holes at $I \setminus I'$", whence a homomorphism

$$\pi^* \longrightarrow \pi'.$$

d3 Let us consider i3 the bases of π, i4 for which d4 d5 the ν' d6 first i6 l_i, i5 are in d7i7 groups with loops d8 $L_{i'}$ $(i' \in I')$ — we call them adapted to I'; they form a torsor $\subset \operatorname{Isom}_{\text{lac}}(\hat{\pi}_{g,\nu}, \pi)$ i8 under the subgroup of d9 $\widehat{\widehat{\mathcal{G}}}_{g;(\nu,\nu')}$ of $\widehat{\widehat{\mathcal{G}}}_{g,\nu} = \operatorname{Aut}_{\text{lac}}(\hat{\pi}_{g,\nu})$ i9 consisting i10 of the $u|h \in \widehat{\widehat{\mathcal{G}}}_{g,\nu}$ whose image in $\mathfrak{S}_I$ leaves invariant d12 the set i11 of the ν' first elements (or, of the $\nu - \nu'$ last its|their complementary set, i12).
m3

Marginal note. m3 Whence also: ∀ basis, ∀ discretification, oriented discretification of π defines one of π', whence ditto for classes of π, π' i13 conjugation, i.e. for|by the strict prediscretifications (possibly oriented) i14, and also for|by the closures of the classes, i.e. the d14 prediscretifications and oriented prediscretifications. It is only the case of arithmetizations|arithmetizations that d15 would require i15 a finer analysis…

For such a basis, we find a corresponding basis of π'. d16 The datum of (π, I') is equivalent to that of a torsor under $\widehat{\widehat{\mathcal{G}}}_{g,(\nu,\nu')}$, that of a π' to d18 the datum of a torsor under $\widehat{\widehat{\mathcal{G}}}_{g,\nu'}$, and the passage from π to π' is described by the change of group of operators

$$\widehat{\widehat{\mathcal{G}}}_{g,(\nu,\nu')} \longrightarrow \widehat{\widehat{\mathcal{G}}}_{g,\nu'}$$

this homomorphism sends $\widehat{\mathcal{G}}_{g,(\nu,\nu')} = \Sigma_{g,(\nu,\nu')}$ into $\widehat{\mathcal{G}}_{g,\nu'} = \Sigma_{g,\nu'}$, [p. 341]
$\cap$
$\Sigma_{g,\nu}$

Apparatus for this scan

140-2-s342-D001 *deletion. Manuscript:* product
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s342-I001 *insertion. Manuscript:* $\xrightarrow{\sim}\Gamma_\Sigma$
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s342-D002 *deletion. Manuscript:* [illeg. 2]
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s342-I002 *insertion. Manuscript:* $\nu' = \operatorname{card} I'$,
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s342-D003 *deletion. Manuscript:* For a map

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

`140-2-s342-I003` *insertion. Manuscript:* Let us consider
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

`140-2-s342-I004` *insertion. Manuscript:* of π,
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

`140-2-s342-D004` *deletion. Manuscript:* $l_{\nu-\nu'+1} \dots l_{\nu}$
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

`140-2-s342-D005` *deletion. Manuscript:* [illeg. 2]
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

`140-2-s342-I005` *insertion. Manuscript:* the ν' last first l_i,
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

`140-2-s342-D006` *deletion. Manuscript:* last
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

`140-2-s342-I006` *insertion. Manuscript:* first
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

`140-2-s342-I007` *insertion. Manuscript:* bases
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

`140-2-s342-D007` *deletion. Manuscript:* bases
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

`140-2-s342-D008` *deletion. Manuscript:* in
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

`140-2-s342-I008` *insertion. Manuscript:* $\subset \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)$
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

`140-2-s342-D009` *deletion. Manuscript:* [illeg. 1]
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

`140-2-s342-D010` *deletion. Manuscript:* +2
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

`140-2-s342-I009` *insertion. Manuscript:* $=$ Aut $\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu})$
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

`140-2-s342-D011` *deletion. Manuscript:* ext
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

`140-2-s342-I010` *insertion. Manuscript:* consisting
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

`140-2-s342-I011` *insertion. Manuscript:* also the set
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

`140-2-s342-D012` *deletion. Manuscript:* also

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s342-D013 *deletion. Manuscript:* +2
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s342-I012 *insertion. Manuscript:* its|their complementary set,
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s342-I013 *insertion. Manuscript:* π, π'
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s342-I014 *insertion. Manuscript:* (possibly oriented)
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s342-D014 *deletion. Manuscript:* [illeg. 1]
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s342-D015 *deletion. Manuscript:* requires
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s342-I015 *insertion. Manuscript:* would require
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s342-D016 *deletion. Manuscript:* the passage
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s342-D017 *deletion. Manuscript:* +2
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s342-D018 *deletion. Manuscript:* [illeg. 1]
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s342-I016 *insertion. Manuscript:* $\overset{\cap}{\Sigma}_{g,\nu}$
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s342-I017 *insertion. Manuscript:* $= \Sigma_{g,(\nu,\nu')}$
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

[scan 343]

and even[i1] $\mathcal{G}_{g,(\nu,\nu')}$ into $\mathcal{G}_{g,\nu'}$, also π into π'. It follows that every prediscretification of π defines one of π', the same for strict prediscretifications, [d1] discretifications, outer bases "adapted to I'" [d2i2]. Finally, if we have a "prearithmetization" of π, i.e. a $\Sigma \subset \widehat{\widehat{\mathcal{T}}}(\pi)$, do we deduce from it a prearithmetization of π' — i.e. (for example)[i3] if two prediscretifications of π are compatible i.e. [d3] have the same group $\Sigma \subset \widehat{\widehat{\mathcal{T}}}(\pi)$, of outer automorphisms, is the same true of their images? [d4]

To see more clearly, we are going to write in diagram form the remarkable sets associated with π, and this in two ways, one without using any particular structure|particular structures on $I = I(\pi)$, the other using a total order, which makes it possible, in the torsor $\mathrm{Isom}_{\mathrm{lac}}(\hat{\pi}_{g,\nu}, \pi)$ under $\widehat{\widehat{\mathcal{G}}}_{g,\nu}$, to define the subtorsor under $\widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu}$ which we may denote [d5] $\mathrm{Isom}^{!}_{\mathrm{lac}}(\hat{\pi}_{g,\nu}, \pi)$. [p. 342]

Apparatus for this scan

140-2-s343-I001 *insertion. Manuscript:* and even

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s343-D001 *deletion. Manuscript:* bases

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s343-I002 *insertion. Manuscript:* "adapted to I'" (i.e.) for which

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s343-D002 *deletion. Manuscript:* (i.e.) for which

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s343-I003 *insertion. Manuscript:* (for example)

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s343-D003 *deletion. Manuscript:* define

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s343-D004 *deletion. Manuscript:* [The question comes down to this:] [the homomorphism

$$\widehat{\widehat{\mathcal{T}}}_{g,(\nu,\nu')} \longrightarrow \widehat{\widehat{\mathcal{T}}}_{g,\nu'},$$

which sends

$$\widehat{\mathcal{T}}_{g,(\nu,\nu')} = \Sigma_{g,(\nu,\nu')}$$

into]

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

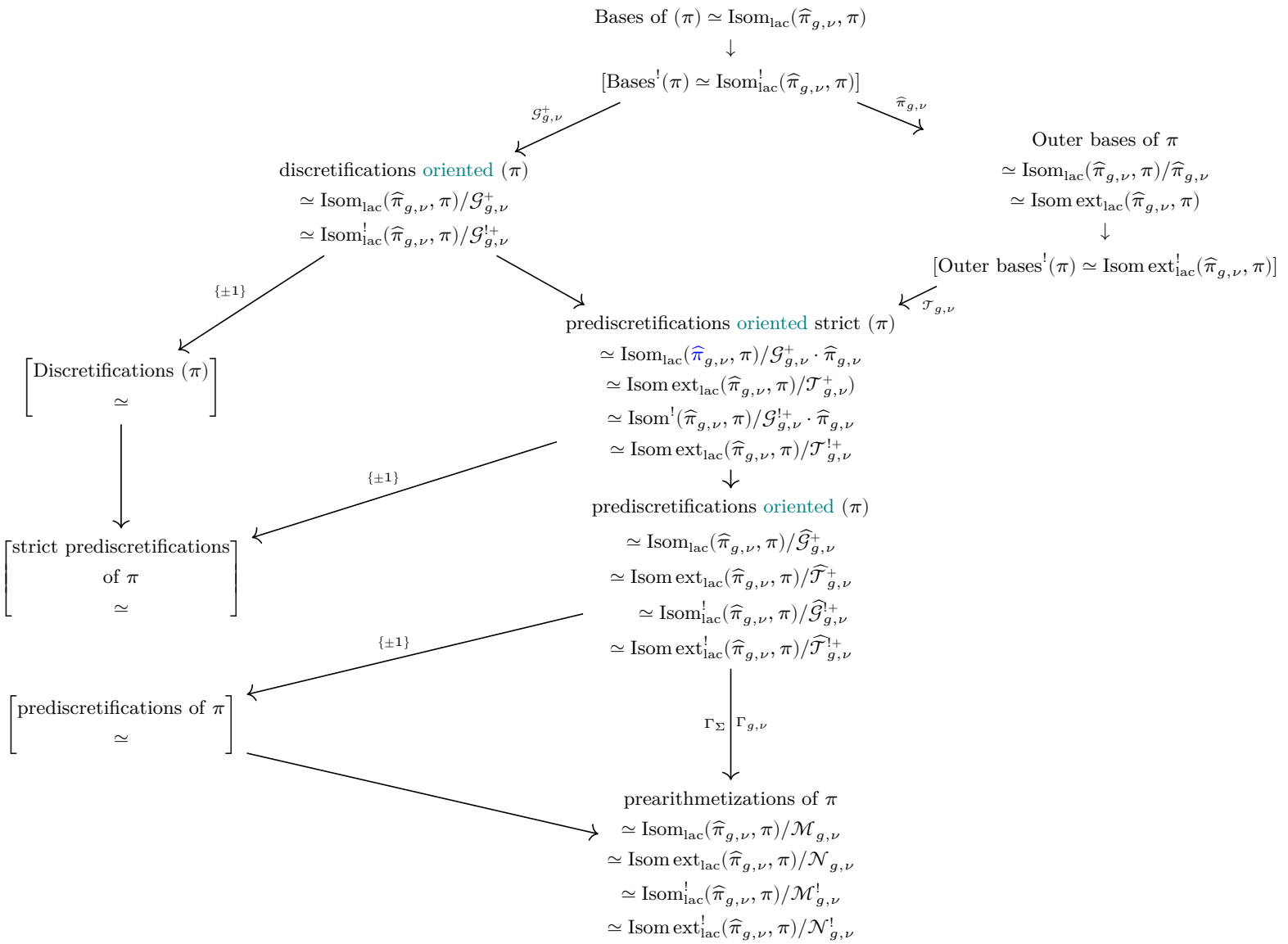
140-2-s343-D005 *deletion. Manuscript:* [illeg. 1]

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

[scan 344]

[p. 174]

[Diagram continued on the landscape page; 140-2-344-D1]


Bases of $(\pi) \simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)$
$[\mathrm{Bases}^{!}(\pi) \simeq \mathrm{Isom}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)]$
$\mathcal{G}^{+}_{g,\nu}$
$\widehat{\pi}_{g,\nu}$
discretifications oriented (π)
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{G}^{+}_{g,\nu}$
$\simeq \mathrm{Isom}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{G}^{!+}_{g,\nu}$
Outer bases of π
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\pi}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)$
$[\mathrm{Outer\ bases}^{!}(\pi) \simeq \mathrm{Isom}\,\mathrm{ext}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)]$
$\{\pm 1\}$
$\mathcal{T}_{g,\nu}$
prediscretifications oriented strict (π)
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{G}^{+}_{g,\nu} \cdot \widehat{\pi}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{T}^{+}_{g,\nu})$
$\simeq \mathrm{Isom}^{!}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{G}^{!+}_{g,\nu} \cdot \widehat{\pi}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{T}^{!+}_{g,\nu}$
Discretifications (π)
$\simeq$
$\{\pm 1\}$
prediscretifications oriented (π)
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\mathcal{G}}^{+}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\mathcal{T}}^{+}_{g,\nu}$
$\simeq \mathrm{Isom}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\mathcal{G}}^{!+}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\widehat{\mathcal{T}}^{!+}_{g,\nu}$
strict prediscretifications of π
$\simeq$
$\{\pm 1\}$
Γ_{Σ}
$\Gamma_{g,\nu}$
prediscretifications of π
$\simeq$
prearithmetizations of π
$\simeq \mathrm{Isom}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{M}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{N}_{g,\nu}$
$\simeq \mathrm{Isom}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{M}^{!}_{g,\nu}$
$\simeq \mathrm{Isom}\,\mathrm{ext}^{!}_{\mathrm{lac}}(\widehat{\pi}_{g,\nu}, \pi)/\mathcal{N}^{!}_{g,\nu}$

m7

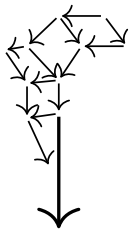
not separated
not separated

Marginal note. m7



m8

Marginal note. m8



[p. 343]

Please turn over

Apparatus for this scan

`140-2-s344-I001` *insertion. Manuscript:* oriented
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

`140-2-s344-D001` *deletion. Manuscript:* ext
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

`140-2-s344-I002` *insertion. Manuscript:* oriented
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

`140-2-s344-I003` *insertion. Manuscript:* oriented
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

`140-2-s344-D002` *deletion. Manuscript:* ≃
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

`140-2-s344-D003` *deletion. Manuscript:*

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

`140-2-344-D1` *Layout placement.* Diagram exceeds the portrait measure after local spacing repair; a landscape page preserves the diagram at 9pt with a marked reference at its original position.

[scan 345]

N.B.

We set [d1]

$$\Gamma_{g,\nu} = \mathcal{N}_{g,\nu}/\Sigma_{g,\nu} = \mathcal{N}^{!}_{g,\nu}/\Sigma^{!}_{g,\nu}$$

(where $\Sigma_{g,\nu} = \widehat{\mathcal{T}}^{+}_{g,\nu}$, $\Sigma^{!}_{g,\nu} = \widehat{\mathcal{T}}^{!+}_{g,\nu}$). We can also describe it as the group Γ_Σ associated with a profinite outer[i1] group with loops π[i2] equipped with an oriented[i3] prediscretification α (nothing more). If we have two such [d2]pairs[i4] $(\pi, \alpha), (\pi', \alpha')$, whence Γ_Σ and $\Gamma_{\Sigma'}$ ($\Sigma = \mathrm{Aut}\,\mathrm{ext}_{\mathrm{lac}}(\pi, \alpha)$, $\Sigma' = \mathrm{Aut}\,\mathrm{ext}_{\mathrm{lac}}(\pi', \alpha')$), we indeed find a canonical isomorphism

$$\Gamma_\Sigma \xrightarrow{\sim} \Gamma_{\Sigma'}$$

by taking any outer isomorphism with loops u[i5] of π with π', transforming α into α' — the associated isomorphism $\Gamma_\Sigma \longrightarrow \Gamma_{\Sigma'}$ obviously does not depend on the choice of u.

This being said, the pairs (π, Σ) of a[i6] profinite|profinite group|groups with loops of type (g, ν), [d3]equipped with a[i7] prearithmetization, with as morphisms the arithmetically outer isomorphisms [d4](relative to[i8] Σ and Σ' ...), form a connected groupoid $\Pi_{g,\nu}$, having an "origin" object (defined up to unique isomorphism as being $(\hat{\pi}_{g,\nu}, \Sigma_{g,\nu})$, or at will any (π, Σ_α) coming from a (π, α) (α an oriented prediscretification of π), and whose group of automorphisms is precisely $\Gamma_{g,\nu}$. [p. 344]

Apparatus for this scan

`140-2-s345-D001` *deletion. Manuscript:* [illeg. 1]

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

`140-2-s345-I001` *insertion. Manuscript:* outer

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

`140-2-s345-I002` *insertion. Manuscript:* π

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

`140-2-s345-I003` *insertion. Manuscript:* oriented

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

`140-2-s345-D002` *deletion. Manuscript:* groups

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

`140-2-s345-I004` *insertion. Manuscript:* pairs

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

`140-2-s345-I005` *insertion. Manuscript:* with loops u

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

`140-2-s345-I006` *insertion. Manuscript:* pairs (π, Σ) of a

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

`140-2-s345-D003` *deletion. Manuscript:* equipped with a

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s345-I007 *insertion. Manuscript:* equipped with a
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s345-D004 *deletion. Manuscript:* transforming
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s345-I008 *insertion. Manuscript:* (relative to
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

[scan 346]

[p. 175]

We observe that, with the exception of the two homogeneous spaces (under $\widehat{\widehat{\mathcal{G}}}(\pi)$, but not necessarily|necessarily under $\widehat{\widehat{\mathcal{G}}}(\pi, I')$, or even under $\widehat{\widehat{\mathcal{G}}}^{!}(\pi)$) Bases (π) and Outer bases (π), the other four are in fact homogeneous spaces under $\widehat{\widehat{\mathcal{G}}}^{!}$, and are expressed as quotients of the $\widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu}$-[i1]torsor $\mathrm{Isom}^{!}(\hat{\pi}_{g,\nu}, \pi)$, by the four subgroups

$$\mathcal{G}^{!}_{g,\nu}, \quad \mathcal{G}^{!}_{g,\nu} \cdot \hat{\pi}_{g,\nu}, \quad \widehat{\mathcal{G}}^{!}_{g,\nu}, \quad \mathcal{M}^{!}_{g,\nu}$$

(the last defined as the inverse image of $\mathcal{N}^{!}_{g,\nu}$ in $\widehat{\widehat{\mathcal{G}}}_{g,\nu}$, just as $\mathcal{M}_{g,\nu}$ is defined as the inverse image of $\mathcal{N}_{g,\nu}$). We find on the other hand, if|for $I' = \{i_0, \dots, i_{\nu'-1}\}$, a|some homomorphism

$$\widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu} \longrightarrow \widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu'},$$

sending $\mathcal{G}^{!}_{g,\nu}$ into $\mathcal{G}^{!}_{g,\nu'}$, [d1] by sending $\pi_{g,\nu}$ into $\pi_{g,\nu'}$, thus $\hat{\pi}_{g,\nu}$ into $\hat{\pi}_{g,\nu'}$, $\widehat{\mathcal{G}}^{!}_{g,\nu} = \Sigma^{!}_{g,\nu}$ into $\widehat{\mathcal{G}}^{!}_{g,\nu'} = \Sigma^{!}_{g,\nu'}$, and the subgroup $\mathcal{G}^{!}_{g,\nu} \cdot \hat{\pi}_{g,\nu}$ of [d2] $\widehat{\mathcal{G}}^{!}_{g,\nu}$[i2] into the subgroup $\mathcal{G}^{!}_{g,\nu'} \cdot \hat{\pi}_{g,\nu'}$ of $\widehat{\mathcal{G}}^{!}_{g,\nu'}$. Finally, since $\Sigma^{!}_{g,\nu}$ maps into $\Sigma^{!}_{g,\nu'}$, [d3] $\mathcal{N}^{!}_{g,\nu}$ maps into the normalizer of $\Sigma^{!}_{g,\nu'}$ (in $\widehat{\widehat{\mathcal{T}}}^{!}_{g,\nu'}$)[i3], [d4]I say that this[i4] is none other than $\mathcal{N}^{!}_{g,\nu'}$. Changing notation, this amounts to the

Lemma The $\mathcal{N}^{!}_{\Sigma} = \mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}^{!}}(\Sigma^{!})$ [d5] $(= \mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}}(\Sigma^{!}) \cap \widehat{\widehat{\mathcal{T}}}^{!})$. not proved [d6]

Obviously we have $\mathcal{N}^{!}_{\Sigma} \subset \mathrm{Norm}_{\widehat{\widehat{\mathcal{T}}}}(\Sigma^{!}) \cap \widehat{\widehat{\mathcal{T}}}^{!}$, conversely let $g \in \widehat{\widehat{\mathcal{T}}}^{!}$ which normalizes $\Sigma^{!}$, let us show that it normalizes Σ i.e. that it is $\in \mathcal{N}$ (thus $\in \mathcal{N}^{!} = \mathcal{N} \cap \widehat{\widehat{\mathcal{T}}}^{!}$). [p. 345]

Apparatus for this scan

140-2-s346-I001 *insertion. Manuscript:* $\widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu}$-
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s346-D001 *deletion. Manuscript:* [illeg. 1] $\widehat{\mathcal{G}}_{g,\nu}$ into $\widehat{\pi}_{g,\nu}$
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s346-D002 *deletion. Manuscript:* [illeg. 1]
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s346-I002 *insertion. Manuscript:* $\widehat{\mathcal{G}}^{!}_{g,\nu}$
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s346-D003 *deletion. Manuscript:* [illeg. 1]

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s346-I003 *insertion. Manuscript:* (in $\widehat{\widehat{\mathcal{T}}}^{!}_{g,\nu'}$)
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s346-D004 *deletion. Manuscript:* which
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s346-I004 *insertion. Manuscript:* I say that this
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s346-D005 *deletion. Manuscript:* [illeg. 1]
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s346-D006 *deletion. Manuscript:* [illeg. 4] of $\mathrm{Norm}_{\widehat{\mathcal{F}}}(\Sigma^{!})/\Sigma^{!}$
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

[scan 347]

It's not clear. I nevertheless[i1] feel like proving|pronouncing [d1] the conjecture The canonical map

$$\text{Prediscretifications|Prediscretifications}^{+}(\pi) \xrightarrow{\sim} \text{Prediscretifications|Prediscretifications}^{+}(\pi')$$

is bijective. In order that[i2] two oriented prediscretifications α, β of π [d2] have the same group of outer automorphisms $\Sigma_\alpha = \Sigma_\beta$, [d3] it is necessary and sufficient [d4] that it be so.[i3] for their images $\Sigma_{\alpha'}$ and $\Sigma_{\beta'}$, i.e. that $\Sigma_{\alpha'} = \Sigma_{\beta'}$; so that the preceding bijection induces a bijection

$$\text{Prearithmetizations}(\pi) \xrightarrow{\sim} \text{Prearithmetizations}(\pi').$$

Finally, [d5] the preceding bijections are compatible with[i4] the homomorphism of the groups of operators[i5] $\widehat{\widehat{\mathcal{T}}}(\pi, I') \longrightarrow \widehat{\widehat{\mathcal{T}}}(\pi')$, [d6] a fortiori[i6] a $\widehat{\widehat{\mathcal{T}}}^{!}(\pi) \longrightarrow \widehat{\widehat{\mathcal{T}}}^{!}(\pi')$, [d7] which implies that it thereby induces|that it induces them (for a given prearithmetization $\Sigma \subset \widehat{\widehat{\mathcal{T}}}(\pi)$[i7] of π, giving $\Sigma' \subset \widehat{\widehat{\mathcal{T}}}(\pi')$[i8] in π'; with $\Sigma^{!} \longrightarrow \Sigma'^{!}$[i9]) a homomorphism $\mathcal{N}^{!}_{\Sigma} \longrightarrow \mathcal{N}^{!}_{\Sigma'}$, [d10] and the corresponding homomorphism

$$\Gamma_\Sigma = \mathcal{N}^{!}_{\Sigma}/\Sigma^{!} \longrightarrow \Gamma_{\Sigma'} = \mathcal{N}^{!}_{\Sigma'}/\Sigma'^{!}$$

is an isomorphism.

To convince oneself of this it suffices to look at [d13] (choosing $\pi = \pi_{g,\nu}|\hat{\pi}_{g,\nu}$, I' [d14] reduced to the|being a last element|element of I_ν)[i10] the case where $I' = \{i'|i\}$, We then have a homomorphism

$$\widehat{\widehat{\mathcal{T}}}(\pi, i) \longrightarrow \widehat{\widehat{\mathcal{G}}}(\pi').$$

(the outer loop automorphisms of π fixing i define well-determined automorphisms of π' — not just outer ones) [p. 346]

Apparatus for this scan

140-2-s347-I001 *insertion. Manuscript:* nevertheless
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s347-D001 *deletion. Manuscript:* [illeg. 1]
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s347-I002 *insertion. Manuscript:* that

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s347-D002 *deletion. Manuscript:* be

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s347-D003 *deletion. Manuscript:* then

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s347-D004 *deletion. Manuscript:* for their images

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s347-I003 *insertion. Manuscript:* that it be so.

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s347-D005 *deletion. Manuscript:* the map $\widehat{\widehat{\mathcal{T}}}(\pi) \longrightarrow \widehat{\widehat{\mathcal{T}}}(\pi')$, which induces

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s347-I004 *insertion. Manuscript:* the preceding bijections are compatible with

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s347-I005 *insertion. Manuscript:* of operators

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s347-D006 *deletion. Manuscript:* which induces

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s347-I006 *insertion. Manuscript:* a fortiori

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s347-D007 *deletion. Manuscript:* this

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s347-I007 *insertion. Manuscript:* $\subset \widehat{\widehat{\mathcal{T}}}(\pi)$

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s347-I008 *insertion. Manuscript:* $\subset \widehat{\widehat{\mathcal{T}}}(\pi')$

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s347-I009 *insertion. Manuscript:* with $\Sigma^! \longrightarrow \Sigma'^!$

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s347-D008 *deletion. Manuscript:* (Σ)

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s347-D009 *deletion. Manuscript:* (Σ')

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s347-D010 *deletion. Manuscript:* which is likewise

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s347-D011 *deletion. Manuscript:* $\widehat{\widehat{\mathcal{T}}}$

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s347-D012 *deletion. Manuscript:* $\mathcal{N}^{!}_{\Sigma'}/\Sigma'^{!}$
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s347-D013 *deletion. Manuscript:* $\pi = \widehat{\pi}_{g,\nu}$
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s347-I010 *insertion. Manuscript:* (choosing $\pi = \pi_{g,\nu}|\widehat{\pi}_{g,\nu}$, I' [illeg. 1] reduced to the|being a last element|element of I_ν)
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s347-D014 *deletion. Manuscript:* [illeg. 1]
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

[scan 348]

[p. 176]

i.e. $\widehat{\widehat{\mathcal{G}}}(\pi, i) \longrightarrow \widehat{\widehat{\mathcal{G}}}(\pi')$ is trivial on $\widehat{\pi}$, and thus passes to the quotient as a $\widehat{\widehat{\mathcal{T}}}(\pi, i) \longrightarrow \widehat{\widehat{\mathcal{G}}}(\pi, i)$. I assume, [d1] by analogy with the discrete case, that this homomorphism is an isomorphism, so that we have an exact sequence [m11]

Marginal note. [m11]

m12

Marginal note. [m12] recall that we assume π anabelian|nonabelian $(2g + \nu \geq 3)$

$$1 \longrightarrow \pi' \longrightarrow \widehat{\widehat{\mathcal{T}}}(\pi, i) \longrightarrow \widehat{\widehat{\mathcal{G}}}(\pi', i) \longrightarrow 1$$

or again

$$1 \longrightarrow \hat{\pi}_{g,\nu-1} \longrightarrow \widehat{\widehat{\mathcal{T}}}_{g,(\nu,\nu-1)} \longrightarrow \widehat{\widehat{\mathcal{G}}}_{g,\nu-1} \longrightarrow 1$$

which contains the exact sequence

$$1 \longrightarrow \pi^{*|+}_{g,\nu-1} \longrightarrow \mathcal{T}_{g,(\nu,\nu-1)} \longrightarrow \mathcal{G}_{g,\nu-1} \longrightarrow 1$$

as|known sub[d2], sequence|is exact. It is convenient to work instead with

$$\begin{array}{ccccccccc}
1 & \longrightarrow & \pi_{g,\nu-1} & \longrightarrow & \mathcal{T}^{!+}_{g,\nu} & \longrightarrow & \mathcal{G}^{!+}_{g,\nu-1} & \longrightarrow & 1 \\
 & & \uparrow & & \uparrow & & \uparrow & & \\
1 & \longrightarrow & \hat{\pi}_{g,\nu-1} & \longrightarrow & \widehat{\mathcal{T}}^{!+}_{g,\nu} & \longrightarrow & \widehat{\mathcal{G}}^{!+}_{g,\nu-1} & \longrightarrow & 1 \\
 & & \| & & \uparrow & & \downarrow & & \\
1 & \longrightarrow & \hat{\pi}_{g,\nu-1} & \longrightarrow & \widehat{\widehat{\mathcal{T}}}^{!}_{g,\nu} & \longrightarrow & \widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu-1} & \longrightarrow & 1
\end{array}$$

The map

[↪ 140-2-348-M1]

$$\text{Prediscretifications}|\text{Prediscretifications}^{+}(\pi) \longrightarrow \text{Prediscretifications}|\text{Prediscretifications}^{+}(\pi')$$

is thus identified with $\widehat{\widehat{\mathcal{T}}}^{!}_{g,\nu}/\widehat{\mathcal{T}}^{!}_{g,\nu}$, we read off the diagram that this is $\simeq \widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu-1}/\widehat{\mathcal{G}}^{!+}_{g,\nu-1}$, whence bijectivity on the sets of oriented prediscretifications.

In the group $\widehat{\widehat{\mathcal{T}}}^{!}_{g,\nu} = \mathcal{T}$, we have an invariant subgroup $\hat{\pi}_{g,\nu-1} = \pi'$, and a subgroup Σ $(= | \longleftarrow \widehat{\mathcal{T}}^{!+}_{g,\nu})$ [p. 347]

Apparatus for this scan

140-2-s348-D001 *deletion. Manuscript:* that
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s348-D002 *deletion. Manuscript:* particular|dense part
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s348-D003 *deletion. Manuscript:* [*illeg.* 1]
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s348-D004 *deletion. Manuscript:* +
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-348-M1 *Layout placement.* The map with long alternative labels is displayed on two lines, broken before its arrow; marker M1 identifies the original running-text position.

[scan 349]

between π' and $\mathcal{T}$, [d1] giving[i1] a subgroup $\Sigma' \simeq \mathcal{T}(\pi')(= \widehat{\mathcal{G}}^{!+}_{g,\nu-1})$[i2] in $\mathcal{T}' = \mathcal{T}/\pi'$ $(\simeq \widehat{\widehat{\mathcal{G}}}^{!}_{g,\nu-1})$, and we know well that then

$$\mathcal{T}/\Sigma \simeq \mathcal{T}'/\Sigma',$$

and [d2] that the normalizer $\mathcal{N}$,[i3] of Σ in $\mathcal{T}$ is the inverse image of the normalizer $\mathcal{N}'$ of Σ' in $\mathcal{T}'$, so that $\mathcal{N}/\Sigma \simeq \mathcal{N}'/\Sigma'$, which proves what we wanted.

Corollary We have canonical isomorphisms

$$\Gamma_{g,\nu} \xrightarrow{\sim} \Gamma_{g,\nu-1} \xrightarrow{\sim} \cdots$$

m13

Marginal note. [m13] We also see that $\Sigma|\Sigma^{!}$ invariant in $\mathcal{T}|\widehat{\widehat{\mathcal{T}}}$ is equivalent to Σ' invariant in $\widehat{\widehat{\mathcal{T}}}'$

so that if $g \geq 2$, we have canonically $\Gamma_{g,\nu} \simeq \Gamma_{g,0}$, on the other hand $\Gamma_{1,\nu} \simeq \Gamma_{1,1}$ $(g = 1, \nu \geq 1)$, $\Gamma_{0,\nu} \simeq \Gamma_{0,3}$ $(\nu \geq 3)$.

N.B. The "fact" [d4] assumed is far from obvious — even that $\widehat{\widehat{\mathcal{T}}}^{!}_{g,(\nu,\nu-1)} \longrightarrow \widehat{\widehat{\mathcal{T}}}_{g,\nu-1}|\widehat{\widehat{\mathcal{M}}}_{g,\nu-1}$ is surjective, or that $\widehat{\widehat{\mathcal{T}}}_{g,(\nu,\nu-1)} \longrightarrow \widehat{\widehat{\mathcal{G}}}_{g,\nu-1}$ is injective, is far from obvious, and is perhaps altogether false !

The assumed fact amounts to a statement of the existence of a functor in the opposite direction "drilling holes", which|what in any case remains not understood.

[p. 348]

Apparatus for this scan

140-2-s349-D001 *deletion. Manuscript:* giving by
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s349-I001 *insertion. Manuscript:* giving
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s349-I002 *insertion. Manuscript:* $(= \widehat{\mathcal{G}}^{1+}_{g,\nu-1})$
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s349-D002 *deletion. Manuscript:* $N(\Sigma)$
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s349-I003 *insertion. Manuscript:* $\mathcal{N}$,
Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s349-D003 *deletion. Manuscript:* $\Gamma_{g,0}$
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s349-D004 *deletion. Manuscript:* [illeg. 1]
Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

[scan 350]

[p. 177]

28. Change of type (g, ν) (continued) : passage to a finite covering.

Let π be a profinite group with loops[i1] of type (g, ν), anabelian as always, π' a subgroup of finite index. We know (or we|we thereby verify, by passing to the discrete case) that [d1] equipped with the[i2] traces on|under π of the loop subgroups of π, π' is a group with loops.

[d2]

The loop subgroups are also|thus the [d6] subgroups L' such that

$$\text{a)} \quad L' = \mathrm{Centr}_\pi(L') \cap \pi'.$$

$$\text{b)} \quad L = | : \mathrm{Centr}_\pi(L') \text{ is a loop subgroup in } \pi.$$

We put $d(L') = [L : L']$, and we thus have a function

$$d : I' = I(\pi') \longrightarrow \mathbb{N}^*,$$

and a map $I' \xrightarrow{\varphi} I$, such that $\forall i \in I$, we have

$$(1) \qquad \sum_{\substack{i' \in I' \\ \text{over } i}} d(i') = n \qquad (\text{where|here } n = [\pi : \pi'])$$

We will have the Hurwitz formula

$$(2) \qquad 2g' - 2 = n(2g-2) + \overbrace{\sum_{i' \in I'} (d(i') - 1)}^{n \,\mathrm{card}\, I - \mathrm{card}\, I'}$$

[p. 349]

Apparatus for this scan

140-2-s350-I001 *insertion. Manuscript:* with loops

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s350-D001 *deletion. Manuscript:* [illeg. 1] discretified

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s350-I002 *insertion. Manuscript:* equipped with the

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.

140-2-s350-D002 *deletion. Manuscript:* If for every $i \in I(\pi)$, we consider an L_i, and the n conjugates $\mathrm{int}(g)L_i = L_{i\alpha}$, with $g \in \pi \bmod \pi'$ $(n = [\pi : \pi'])$, n $1 \leq \alpha \leq n)$ counted|to count the number of times that a conjugacy class in π' of the subgroup $\mathrm{int}(g)L_i$ is then if $d_\alpha | d_{i\alpha} = [L_{i\alpha} : L'_{i\alpha} | N'_{i\alpha} \cap \pi']$ for $L'_{i\alpha} = L_{i\alpha} \cap \pi'$, then exactly $d_{i\alpha}$ among the $L'_{i\beta}$ $(1 \leq \beta \leq n)$ are conjugate in π' to $L'_{i\alpha}$, and if we sum over the distinct conjugacy classes of the subgroups

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s350-D006 *deletion. Manuscript:* the

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s350-D007 *deletion. Manuscript:* $2g|2\alpha$

Text: omitted Reason: Cancelled in the manuscript: omission follows the author's final state; uncertainty in the cancelled material remains marked.

140-2-s350-I007 *insertion. Manuscript:* $n \operatorname{card} I - \operatorname{card} I'$

Text: insertion integrated Reason: Explicit authorial insertion: integrated at the position recorded by the transcription, without changing its reading.